Demonstrating the Limitations of Neural Networks in Data-Limited Materials Discovery

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Abstract

Neural networks are increasingly applied to materials discovery, where they are tasked with predicting the properties of structures and compositions that have yet to be constructed physically. Despite their theoretical capacity to approximate any continuous function, neural networks are fundamentally interpolative: their predictions are reliable only within the boundaries of the data on which they were trained. This presents a fundamental conflict with the extrapolative demands of scientific discovery.

We investigate this conflict through a series of controlled experiments on synthetic datasets with known ground-truth functions, isolating specific failure modes including extrapolation beyond the training distribution, failure in regions of missing data coverage, and overfitting in the presence of noise. We further demonstrate that the choice of activation function, an architectural decision made prior to training, influences extrapolative behavior independently of model performance within the training region. These findings are then contextualized in a real materials system, where a neural network trained on glass compositions at low modifier concentrations is evaluated at higher concentrations never seen during training. The resulting predictions diverge from both theoretical and experimental reference data, consistent with the behavior observed in the synthetic experiments.

Taken together, these results motivate a more critical approach to the deployment of neural networks in scientific applications, and highlight the value of physically grounded and interpretable alternatives.

1 Introduction

The discovery of new materials is an inherently predictive enterprise. Identifying promising compositions requires making informed judgments about systems that have not yet been constructed or tested, relying on patterns learned from existing experimental data to anticipate the behavior of novel ones. The cost and time associated with physical synthesis make accurate prediction especially valuable: a model that can reliably estimate the properties of unsynthesized compositions could dramatically accelerate the discovery process.

Neural networks have emerged as a natural candidate for this role. As universal function approximators [1], they are capable (in principle) of learning arbitrarily complex relationships from data. Their flexibility and scalability have made them an increasingly common tool in computational materials science, where they have been applied to tasks ranging from property prediction to the identification of stable phases [2].

However, this flexibility comes with an important caveat. Neural networks learn by recognizing patterns within their training data, with no explicit architecture that grounds these predictions in physical truth. Formally, they are interpolators: when queried at inputs that lie outside or between the clusters of their training distribution, their outputs are not constrained by the underlying function and can deviate arbitrarily [3, 4]. This property stands in direct conflict with the demands of materials discovery, where the inputs of greatest interest are precisely those that lie outside the training distribution.

This work examines that conflict directly. We present a series of experiments on synthetic datasets, chosen to isolate specific failure modes in a controlled setting. We then apply the same analysis to a real glass-forming system, demonstrating that the failures observed synthetically manifest in physically meaningful ways. Our aim is not to dismiss neural networks as a tool, but to characterize the conditions under which they should be trusted, and to motivate the use of physically grounded alternatives where those conditions are not met.

2 Background

2.1 Neural Network Architecture

A feedforward neural network maps an input vector to an output vector through a sequence of learned transformations. The network is organized into layers: an input layer, one or more hidden layers, and an output layer. Each neuron in a hidden layer computes a weighted sum of its inputs and passes the result through a nonlinear activation function, introducing the expressive capacity that distinguishes neural networks from linear models.

The choice of activation function has meaningful consequences for model behavior. Three functions are considered in this work: the rectified linear unit (ReLU), $f(x) = \max(0, x)$, is piecewise linear and unbounded. The hyperbolic tangent, $f(x) = \tanh(x)$, is bounded on $(-1, 1)$ and saturates at its asymptotes. The sigmoid, $f(x) = (1 + e^{-x})^{-1}$, is bounded on $(0, 1)$ and similarly saturates. These asymptotic properties become significant when a network is queried outside its training range.

2.2 The Interpolation-Extrapolation Distinction

A model *interpolates* when a query point falls within the support of its training distribution, and *extrapolates* when it does not. In one dimension, this reduces to a simple boundary: a model trained on $x \in [a, b]$ interpolates within that interval and extrapolates beyond it. Neural networks are not explicitly constrained to produce reasonable outputs in the extrapolative regime, and their behavior there is determined by architecture rather than data [3, 4].

2.3 Glass Structure Prediction

The structural properties of oxide glasses are commonly characterized by the distribution of Q^n species, where n denotes the number of bridging oxygen bonds at a given network-forming site. This distribution varies continuously with composition and is measurable experimentally. It has been demonstrated that a statistical mechanics (SM) model can predict Q^n distributions from an energetic perspective, and that augmenting a neural network's input features with SM model outputs improves predictive accuracy by grounding the network's learned representation in physical theory. This system serves as the applied context for the present work [2].

3 Experimental Framework

3.1 Synthetic Datasets

Experiments are conducted on three functional forms: linear ($y = ax + b$), polynomial ($y = x^2$), and sinusoidal ($y = \sin(x)$). Datasets are generated by thoroughly sampling over a specified range, evaluating the true function, and optionally adding Gaussian noise. Because the underlying function is known exactly, model predictions can be evaluated not just against held-out data points but against the true relationship everywhere.

3.2 Splitting Strategies

Two splitting strategies are used. A *random split* draws training and test points from the same distribution, placing the model in an interpolative setting. A *range-based split* restricts training data to a contiguous sub-interval of the full input range, with test data extending beyond it, placing the model in an extrapolative setting.

3.3 Model Architecture and Training

A fixed architecture is used across all experiments: a feedforward network with two hidden layers of 32 neurons each, trained with the Adam optimizer and mean squared error loss. Activation functions are varied by experiment. Holding architecture constant isolates the effect of the variable under study — activation function, split strategy, or dataset — from confounding architectural differences.

4 Results

4.1 Activation Functions and Extrapolation

Three networks are trained identically on a linear function over $x \in [0, 2]$ and evaluated over $x \in [0, 4]$, with activation functions varied across ReLU, tanh, and sigmoid. Within the training range, all three converge to comparable performance. Beyond it, their behavior diverges markedly (Figure 1). ReLU, being piecewise linear, continues with a fixed slope. Tanh and sigmoid, bounded by construction, saturate and fail to track any upward trend. The extrapolation behavior of each model is determined entirely by its activation function — an architectural choice made before training begins.

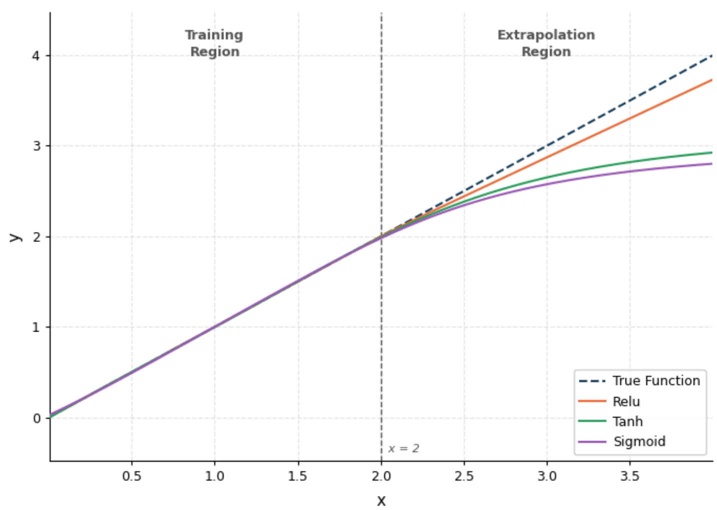

Figure 1: Three networks trained on a linear function in $[0, 2]$ diverge beyond the training boundary. ReLU continues linearly; tanh and sigmoid saturate at their asymptotes. The true function is shown as a dashed line.

4.2 Interpolation and Extrapolation

A network trained on $\sin(x)$ over $[0, 2\pi]$ is evaluated over $[0, 4\pi]$. The model fits the training region well but fails to recover the periodic pattern beyond the training boundary (Figure 2). The network has learned a representation adequate for the data it was shown, but has acquired no knowledge of the underlying function.

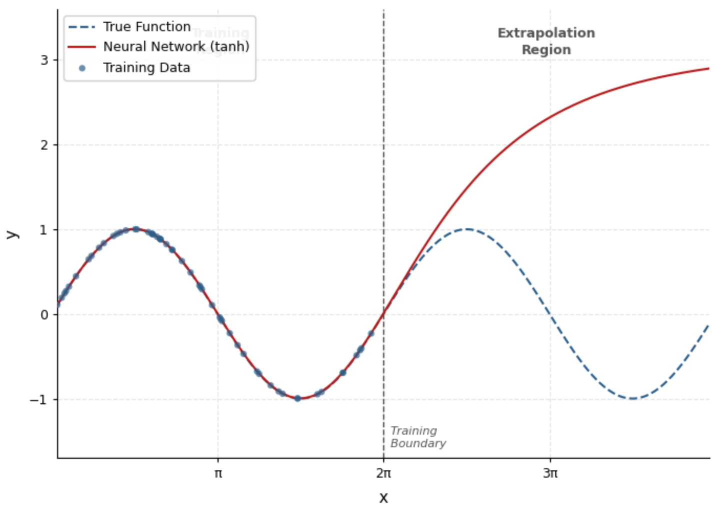

Figure 2: A network trained on $\sin(x)$ over $[0, 2\pi]$ fails to continue the pattern into $[2\pi, 4\pi]$. Strong performance within the training region does not imply reliable extrapolation beyond it.

This failure is partly structural. The activation functions used here — ReLU, tanh, and sigmoid — are not periodic, and a network composed of them cannot produce periodic outputs regardless of its depth or width. The extrapolation behavior observed in Figure 2 is therefore not merely a consequence of insufficient training data; it reflects a fundamental mismatch between the representational capacity of the architecture and the symmetry of the target function. Periodic activation functions, such as $\sin(x)$, have been proposed as an alternative [6] and can in principle support periodic extrapolation, but they introduce their own training difficulties and are not standard. The broader point stands: extrapolative behavior is constrained by architectural choices made before training begins, and no amount of data can overcome a mismatch between model structure and the structure of the underlying function.

4.3 Data Coverage

A network is trained on $y = x^2$ using data drawn from $[-2, -1] \cup [1, 2]$, with no observations in $[-1, 1]$. Despite having seen both arms of the parabola, the model fails entirely in the unobserved interior (Figure 3). Adequate coverage of the input space is a necessary condition for reliable prediction, and cannot be inferred from the range of training inputs alone.

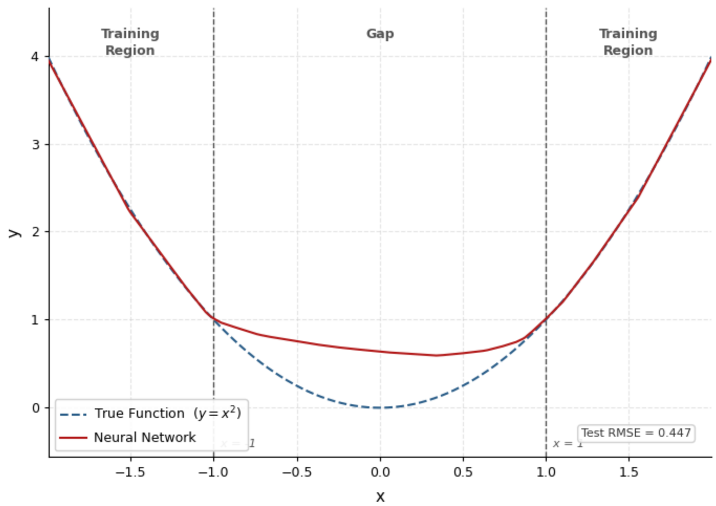

Figure 3: A network trained on x^2 outside $[-1, 1]$ fails in the unobserved gap, despite the true function being smooth and regular throughout. Boundary lines mark the edges of the training regions.

4.4 Overfitting and Noise Sensitivity

Two networks are trained on $\sin(x)$ with 100 observations: one on clean data, one with added Gaussian noise ($\sigma = 0.2$). Both are trained for 20,000 epochs. On clean data the model fits the underlying function closely. On noisy data it fits the noise, producing a prediction that tracks individual observations rather than the trend they represent (Figure 4). This behavior is particularly consequential in data-limited settings, where the model has no basis for distinguishing signal from noise.

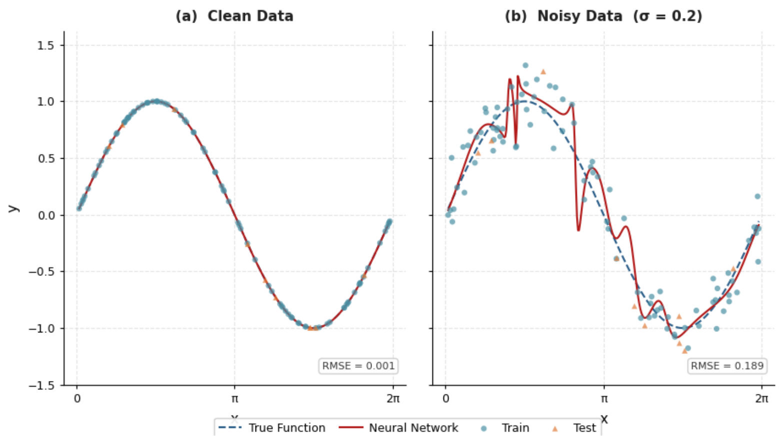

Figure 4: Left: a network trained on clean sinusoidal data fits the underlying function. Right: the same network trained on noisy data overfits to individual observations. Training data are shown as points; the true function is dashed.

5 Application to Glass Structure Prediction

The synthetic experiments establish failure modes in a controlled setting. We now examine whether the same behaviors arise in a real materials system. A network is trained on glass compositions drawn from the full multicomponent dataset, restricted to those with sodium concentrations below 33 mol%. It is then evaluated across the complete binary NaSi composition range, from 0 to 66 mol% Na. This reproduces the range-based split of Section 3 on physically meaningful data, where the true structural behavior is independently known from both statistical mechanics theory and experiment.

Figure 5 shows the predicted Q^n species distributions alongside the theoretical SM model and experimental measurements. Within the training region, the neural network tracks both references closely. Beyond the training boundary, predictions diverge from the SM model and fail to follow the experimental trend. The character of the divergence — a gradual decoupling from the true function with increasing distance from the boundary — is consistent with the behavior observed in the synthetic extrapolation experiments.

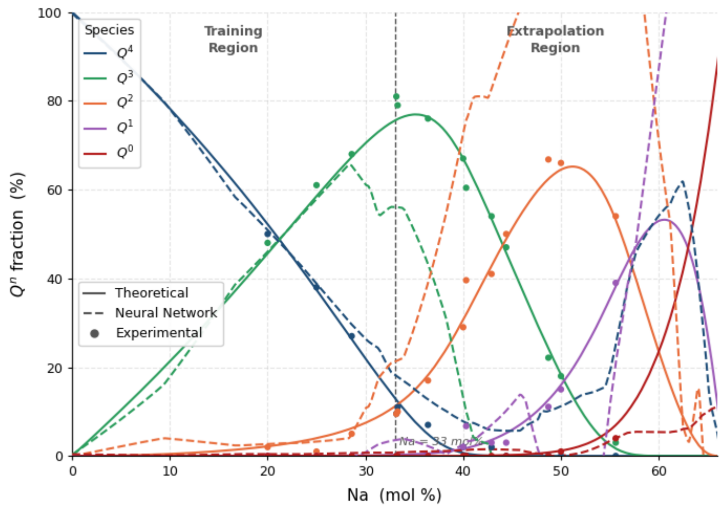

Figure 5: Predicted Q^n fractions across the NaSi composition range. The network is trained on compositions with Na < 33 mol% and evaluated across the full range. Solid lines show the theoretical SM model; dashed lines show neural network predictions; markers show experimental data. A dashed vertical line marks the training boundary; predictions diverge from theory and experiment beyond it.

6 Discussion

6.1 Synthetic and Applied Results

The failure observed in the NaSi system is consistent with those established synthetically: reliable performance within the training distribution, and systematic degradation beyond it. This consistency is itself informative. The synthetic experiments isolate failure modes under controlled conditions; the NaSi result demonstrates that these modes are structural properties of neural networks, not artifacts of simplified data. A model trained on physically real compositions, with a physically real target, exhibits the same interpolative boundary as a model trained on $\sin(x)$.

6.2 Limitations Beyond Predictive Accuracy

The experiments in this work evaluate performance through RMSE, which captures the magnitude of prediction error but not its nature. Three further limitations are relevant to scientific applications and are not reflected in this metric.

Silent failure: A standard feedforward network produces a point prediction with no accompanying uncertainty estimate. Within the training distribution and beyond it, outputs are numerically indistinguishable in form. A researcher has no signal from the model itself that a query lies outside the region where predictions can be trusted.

Interpretability: The learned representation of a neural network — its weights and activations — does not correspond to any physical quantity. A prediction cannot be examined to understand why it was produced, nor can it be connected to a mechanistic account of the system being modeled.

Absence of functional form: Physical laws are expressed as equations. A neural network does not produce one. Its predictions cannot be verified for physical plausibility, incorporated into theoretical frameworks, or used to derive relationships that generalize beyond the training data. This distinguishes neural networks fundamentally from methods that search the space of mathematical expressions directly.

7 Outlook

The limitations identified in this work are not unique to the architecture or dataset studied. They reflect properties common to feedforward neural networks broadly, and addressing them requires either constraining the model with external knowledge or replacing it with a method better suited to the task.

Physics-informed machine learning. One approach is to incorporate domain knowledge directly into the model. [2] demonstrate this by augmenting neural network inputs with outputs from a statistical mechanics model, anchoring predictions in physically grounded quantities and reducing the burden placed on the network to extrapolate from composition alone. This does not eliminate the interpolative boundary, but narrows the space in which extrapolation is required.

Training data augmentation. A complementary approach addresses the extrapolation problem without modifying the model or requiring domain knowledge. [5] propose generating synthetic training samples at positions just outside the training boundary, using the network's own predictions to iteratively extend its reliable range outward. This targets the extrapolation problem directly at the data level, and is applicable in settings where neither physical priors nor interpretable outputs are required.

Symbolic regression. Where the goal is to recover an underlying functional relationship rather than simply predict outputs, methods that search the space of mathematical expressions directly are more appropriate. Genetic algorithm-based approaches such as PySR [7] return explicit equations that can be examined, verified against physical theory, and generalized beyond the training data. For scientific applications where interpretability and physical plausibility are requirements, this represents a more natural fit than a neural network.

Neither approach is without limitations of its own, and the right choice depends on the structure of the problem, the availability of prior knowledge, and the degree to which interpretability is required. The contribution of this work is not to prescribe a replacement, but to clarify the conditions under which a replacement should be sought.

8 Conclusions

Neural networks are powerful interpolators. Within a well-sampled training distribution they can approximate complex functions with high accuracy, and their flexibility makes them applicable across a wide range of tasks. The experiments presented here do not challenge this. What they demonstrate is that this capability does not extend reliably beyond the training boundary, that the manner in which it fails is determined by architecture rather than data, and that standard evaluation metrics provide no warning when it does.

These properties are consequential for materials discovery, where the most valuable predictions concern compositions and conditions that have not been measured. A model that performs well on held-out data drawn from the same distribution as its training set may still fail systematically when deployed on genuinely novel inputs. The NaSi experiment illustrates this directly: strong in-distribution performance does not survive the extrapolative demands of the application.

The appropriate response is not to abandon neural networks, but to deploy them with a clear account of what they can and cannot do. Where extrapolation is required, where data is limited, or where physical interpretability matters, the choice of model deserves the same scrutiny as any other experimental decision.

References

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Code: `https://github.com/atpower89/Neural-Network-Project`